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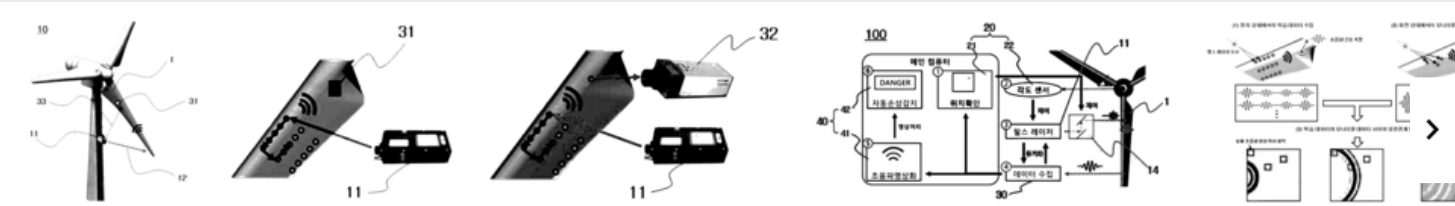
Laser ultrasonic imaging method and laser ultrasonic imaging device for rotational structure
WO 2014021564 A1

ABSTRACT

The present invention relates to an integrity evaluation system which is capable of visualizing the interaction between a damaged portion and an ultrasonic procedure such that field workers can confirm information on the location and the size or the like of the damage in an easy and intuitive manner, thereby efficiently managing a rotational structure such as a turbine blade, and the integrity evaluation system comprises: an ultrasonic generation system which irradiates pulse laser on one side of the rotational structure so as to generate ultrasonic waves; a pulse laser control system for adjusting a time for irradiating the pulse laser; an ultrasonic measurement system for measuring ultrasonic waves generated from the other side of the rotational structure; and a damage detection system which visualizes a measured ultrasonic signal and automatically provides information on the existence of damage and the position and severity of the damage of the rotational structure.

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DESCRIPTION translated from Korean

Laser ultrasonic imaging method and apparatus of the rotating structure

The present invention relates to a laser ultrasonic imaging method and apparatus of the rotating structure, and more particularly, to turbine blades caused damage to the various local safety intuitive diagnosis using laser ultrasound-based health monitoring system for in (turbine blade) It relates to a method and apparatus for performing.

The present invention supports the future creation of Sciences, National Research Foundation of Korea funded (MSIP) (NRF) (Report Number: 20,110,016,470) and the support of the Ministry of Industry usually funded Korea Institute of Energy Technology Evaluation and Planning (MOTIE) (KETEP) (Report Number: 20,123,030,020,010) is the result performed with the.

Excessive dependence on fossil fuels is coming up the recent worldwide energy crisis. Nuclear power has received attention as a solution to overcome these fossil fuel dependence, including the Chernobyl nuclear power plant accident in Fukushima nuclear power plant explosion in Japan recently seen in accident and cool Marquez won the French nuclear waste treatment facilities flyers explosion as concerns about their safety It is increased.

As an alternative, the international solar power, hydrogen energy, such as wind power has been devoting the energies on the development of green energy sources. In particular, the power generation cost of wind power is cheaper and

CLAIMS (20) translated from Korean

1. In the rotating structure integrity assessment system,

Ultrasound generating system for generating ultrasonic waves by irradiating a pulse laser to a side of the rotary structure;

Pulsed laser control system for adjusting the irradiation time of the laser pulse;

Ultrasonic measuring system for measuring the ultrasonic wave generated from the other side of the rotary structure; And

Imaging the measurement of the ultrasonic signal and the rotation comprises a damage detection system that automatically provides for damage, damage location and severity of the rotary structure information structure integrity evaluation system
2. The method of claim 1,

The ultrasonic measuring system rotating structure quality assessment system comprising: a digitizer for collecting and storing the ultrasonic sensor and the signal from the sensor for detecting the ultrasonic signal
3. 3. The method of claim 2,

attracts a large-scale development of this new alternative energy sources possible.

In the construction of wind turbines around the world, but actively going along, regular research on health diagnostic techniques that can detect damage and ensure the reliability of the wind turbine situation is still insufficient. In particular a wind turbine blade is damaged due to lifting off the external object, tends to fatigue and damage by lightning strikes or the like due to a blast, early detection of damage is critical to the efficient development and maintain structural integrity.

Conventional rotating blade health monitoring and diagnostic techniques have been proposed optical fiber sensor-based monitoring is non-patent munheo 1. The method presented in Non-Patent munheo 1 to measure the strain applied to the load change and the like of the optical fiber sensor is installed in position with that in the event of strain on the optical fiber sensor according to the wavelength of the reflected light is changed, such as the length of the grating is changed to analyze. However, calculating the strains, such as information from the measurement wavelength changes and complex data analysis process is required in order to diagnose the damage, the production manager for inserting an optical fiber inside the blade from the blade manufacturing process is required.

Non-Patent Document 2 is a wind turbine blade monitoring method using an acoustic sensor, such as impact damage or injury when the crack occurs, the sound wave due to the stress / strain changes in the structure are released, by measuring this time the damage caused by the sensor that the presence a can be immediately detected, such as an analysis of the shape of the measured acoustic wave arrival times and the sound waves emitted (corruption occurs) position can also be checked. However, the signal is weak, the signal-to-noise ratio is high, the interpretation difficult, the degree of freedom compared to the other active monitoring techniques with passive monitoring technique capable of measuring a change occurs only relatively poor.

Conventional techniques are inferior in the cable installation costs, and maintenance costs due to a durability problem, due to the limited number of sensors due to the limitation of high-density sensor network construction for these real-world applicability because it requires the installation of driving a plurality of sensors.

Therefore, the helicopter blades of a wind turbine, including the aircraft and monitoring systems for the rotational structure of the vessel, such as propeller engine technology that enables the measurement of the wind generator while reducing the number of required high-density and high-resolution built-in sensor is required, and practitioners in the field Intuitive health monitoring system is to be interpreted as corruption is also needed to operate the system easily.

[Prior Art Document]

(Non-Patent Document 1) Schroeder, K., et al. "A fibre Bragg grating sensor system monitors operational load in a wind turbine rotor blade," Meas. Sci. Technol. 17, 1167 (2006).

(Non-Patent Document 2) Blanch, MJ and Dutton, AG, "Acoustic emission monitoring of field tests of an operating wind turbine," Key Eng. Mater. 245-246, 475-482 (2003).

The present invention has been made in view of the above, depending on the need, for a variety of rapid local damage occurring in the rotating blades and the exact field applicability for performing a safety check is straightforward to provide a high quality at all times monitoring system.

Configuration of the quality evaluation system of the rotary structure according to the invention for the problem to be solved by the above, by irradiating a pulse laser to a side of the rotating structure to generate the ultrasonic wave system generates ultrasound, pulse for adjusting the irradiation time of the laser pulses The laser control system, the ultrasound measuring system and measuring the

The ultrasonic sensor rotary structure integrity evaluation system, characterized in that the integrated structure of the rotation sensor provided in the other side of the laser irradiation point

4. 4. The method of claim 3,

The rotation sensor integrated structure integrity evaluation system, characterized in that the piezoelectric sensor sensitivity

5. 3. The method of claim 2,

The ultrasonic sensor rotary structure integrity evaluation system, characterized in that the piezoelectric sensor node of wireless

6. 3. The method of claim 2,

The ultrasonic sensor rotary structure integrity evaluation system characterized in that the non-contact laser interferometer

7. The method of claim 1,

Rotating structure integrity evaluation system comprising the ultrasound generation system Nd-YAG pulsed laser

8. The method of claim 1,

The ultrasound generating system rotating structure integrity evaluation system further comprises a galvanometer in order to adjust the direction of the target to the laser beam direction

9. The method of claim 1,

The pulsed laser control system rotating structure integrity evaluation system synchronizes the laser pulses with the blades, characterized in that to install the angle sensor to the rotary shaft structure in order to irradiate the laser beam only enter in the target range

10. The method of claim 1,

The pulsed laser control system rotating structure integrity evaluation system, characterized in that to generate an electrical pulse signal each time the blade is rotated once with an encoder for detecting an initial position

11. The method of claim 1,

The damage detection system is damaged or without ultrasound image processing unit and image processing to generate an ultrasonic wave, rotating the location and severity of damage to the information being configured to automatically detect damaged parts automatically to provide structural integrity evaluation system

12. A pulse laser irradiation step on one single point of the structure in a state where the rotary structure and stationary collecting training data to collect ultrasound generated using the built-in sensor and a laser oscillating system;

Collecting the structure is generated by the irradiation of a laser beam while rotating the ultrasonic sensor through the data acquisition and monitoring and storing; And

imaging ultrasonic signals to measure the ultrasonic wave generated from the other side of the rotary structure comprises a damage detection system which provides for damage, damage location and severity of the rotary structure information automatically .

In a preferred embodiment of the invention, the ultrasonic wave generating system Nd-YAG pulsed laser to adjust the laser beam in a direction outside the target direction, and is characterized in that it comprises a galvanometer for the energy density adjusted.

In a preferred embodiment of the invention, the ultrasonic measurement system includes a digitizer to collect and store the signal from the ultrasonic sensor and the sensor for detecting the ultrasonic signal, the ultrasonic sensor is installed on the rotating structure on the other side of the laser irradiation point Built-in high-sensitivity sensor and a piezoelectric sensor or laser interferometer is characterized in that the piezoelectric sensor node wireless or contactless instrument.

In a preferred embodiment of the invention, the pulsed laser control system is such that the angle sensor (encoder), the rotation shaft structure in order to irradiate the laser beam only if the blade comes within the target range to synchronize with the pulse laser, a rotating blade is accumulated to generate an electrical pulse signal every time characterized in that for detecting the initial position.

In a preferred embodiment of the present invention, the damage detection system and the ultrasound image processing unit for image processing or without damage to the ultrasonic wave generation, and the location and the severity of the damage information, characterized by automatically sensing that provides damage-part automatically.

In another embodiment of the present invention, the laser ultrasonic imaging method, the rotation structure of the pulsed laser irradiation from a single point on one side of the stationary structure and built-in sensor and a learning data collection step of collecting the generated ultrasonic waves by means of a laser interferometer ;

The collection structure is generated by the irradiation of a laser beam while rotating the ultrasonic sensor or a laser interferometer, and through the integrated monitoring and storing the data collected; And

Analyzing the correlation between the training data and the monitoring data to estimate the location and ultrasound comprising the step of imaging the image.

In a preferred embodiment of the invention, the learning step in a data collection point before learning repeatedly irradiated by the scanning laser over an area comprising the following steps to create the training data set was collected ultrasound generated for all the learning point, and The monitor in the data collection step, and repeatedly to the scanning laser irradiation in the monitoring area, and then by collecting the generated ultrasonic waves to all the monitoring area, and further comprising the steps of creating a monitor data set, the training data and the correlation index of the monitoring data It characterized in that the ultrasonic generator to identify the point when the maximum.

In another embodiment of the invention, the quality evaluation method for a rotary structure, irradiating a laser spot on one side of the rotation number of the structure, and measuring the ultrasonic wave emitted from the other fixed point of the rotating structure, the measured ultrasonic comprising: using the opposite principle of the ultrasonic imaging process data, in order to emphasize the damaged region using a standing wave filter calculates the value of the standing wave trapped in the wave energy and the damaged areas to visualize the damaged step and compared with the reference value of the damage presence and automatically is characterized in that it includes the step of providing information on the location and severity.

Laser ultrasound imaging method, characterized in that for estimating the position comprises the step of imaging ultrasound image by analyzing a correlation between the training data and the monitoring data,

13. 13. The method of claim 12,

Laser ultrasound imaging method, characterized in that the learning step in a data collection point before learning repeatedly irradiated by the scanning laser over the area, and then collects the ultrasound generated with respect to all the learning point, further comprising the step of creating a set of training data

14. 13. The method of claim 12,

The monitoring on the data collection step, and repeatedly scanning the laser irradiated area, and then a laser ultrasound imaging method for monitoring and collecting ultrasound generated for all the steps to create a monitoring area monitored data set further comprises

15. 13. The method of claim 12,

Laser ultrasound imaging methods, the correlation index of the learning data and the monitoring data, characterized in that the ultrasonic generator to identify the point when the maximum

16. In prudential assessment methods for rotating structure,

Irradiating a laser spot on one side of the rotation number of the structure;

Measuring at one point, fixed to the other side of the generated ultrasonic rotary structure;

Comprising: an ultrasonic imaging process using the opposite principle of the measured ultrasound data;

Step of visualizing the injury using a standing wave filter in order to emphasize the damaged area; And

Calculating the energy value of the standing wave wave trapped in the damaged area, and quality evaluation method for a rotary structure compared with a reference value, it characterized in that it comprises the step of providing information about the presence and location and severity of damage to the existing automatic

17. The blades of the rotary structure;

And generating an ultrasonic pulse laser is irradiated with a laser beam to one side of the blade

Rotating structure damage monitoring system comprising: an ultrasonic sensor for detecting ultrasonic waves generated from the other side of the blade

18. 18. The method of claim 17,

The ultrasonic sensor rotary structure damage monitoring system characterized in that the piezoelectric sensor is attached to the blade

19. 18. The method of claim 17,

The ultrasonic sensor rotary structure damage monitoring system characterized in that the piezo-electric sensor

In another embodiment of the invention, the configuration of the rotary structure damage monitoring systems, the blades of the rotary structure, a laser beam to one side of the blade and to detect ultrasonic waves generated by the pulse laser and the other side of the blade of generating ultrasonic waves and a ultrasonic sensor.

In a preferred embodiment of the invention, the ultrasonic sensor is characterized in that it is a wireless or non-contact laser interferometer piezoelectric sensor attached to the piezoelectric sensor, and the blade attached to the blade.

The present invention provides an automatic on-site information about the turbine blades, helicopter, aircraft, and marine engines, such as the location of the damage of the propeller to visualize the interaction between the damage and the ultrasonic process for various rotational structure, size and severity of the wind turbine. It enables practitioners to the effective management by allowing you to also find easy and intuitive.

In addition, or can be rapid diagnosis at an early stage of the damage to the small size possible damage to the sensitive detection of damage to carry out detection process based on the ultrasound image with high resolution using the laser scanning sensors and the installation requires less than conventional techniques is required.

In addition, it is possible to perform efficient and reliable quality assessed by running without interruption during operation of the rotating structure, as well as possible, the real-time diagnostics to determine the damage includes an algorithm automatically.

Through this approach, such as offshore wind farms can reduce maintenance costs by performing remote diagnostics efficiently even in tough safety rotating structure.

1 is generated ultrasonic blades for wind turbines according to the present invention the system

Figure 2 is an ultrasonic measuring system with a built-in sensor according to an embodiment of the present invention

3 is an ultrasonic measuring system with a laser oscillating system according to an embodiment of the present invention

Figure 4 is the structure of a single integrated quality evaluation system according to an embodiment of the present invention

Figure 5 is an imaging technique that shock localization according to the invention

Figure 6 is an example experiment of the present invention

Figure 7 is a photograph of an ultrasound image processing result to the invention

Figure 8 is a picture visualization by extracting only damage the imaging information as the damage visualization technique in accordance with the present invention

Rotating structure integrity evaluation system according to the present invention, by irradiating a pulse laser to a side of the rotating structure to generate the ultrasonic wave system generates ultrasound, pulsed laser control system for adjusting the irradiation time of the laser pulse, generated from the other side of the rotary structure. The ultrasonic measurement system and measuring the imaging ultrasound to measure the ultrasonic signal, and characterized in that it comprises a damage detection system that provides yomu damage, injury severity and location information of the rotary structure automatically.

Hereinafter, a preferred embodiment with reference to the accompanying drawings, which can be easily subjected to self invention having ordinary skill in the art in detail. In the drawings of the present invention, the size and dimensions of the structures is a diagram to enlarge or reduce than actual for clarity of the present invention, a characteristic configuration of a known configuration and does not limit are omitted to reveal a view showing hayeoteumeuro. In this case a detailed description of known functions and configurations in the following description of the details of the principle relating to the preferred embodiment of the present invention is determined to unnecessarily obscure the subject matter of the present invention, a detailed description thereof will be omitted.

Figure 1 shows an ultrasonic generation for health monitoring of the blade of a wind turbine according to an exemplary embodiment of the rotary structure according to the present invention. If the irradiation of pulse laser 11 to the laser beam 12 to a side of the blade (1) is generated in the ultrasonic blade (1). 1 is shown with respect to the blade would (1) of a wind turbine, without limited to the helicopter, the propeller of the aircraft and a ship engine and the like can be applied to a variety of rotating structures. Rotation structure of materials such as aluminum or steel or metal CFRP (Carbon Fiber Reinforced Plastic) or GFRP (Glass Fiber Reinforced Plastic), such as composite materials, such as ultrasonic waves can be generated in any respect material to the application you want to be there.

Pulse laser 11 is to raise the temperature of the irradiated area has got a high energy locally, where the thermal expansion due to the thermal energy is propagated to the ultrasonic type. As a pulse laser is irradiated to the high energy level it will

attached to the radio to the blade

20. 18. The method of claim 17,

The ultrasonic sensor rotary structure damage monitoring system characterized in that the non-contact laser interferometer

not yungbal (ablation) of the blade surface, preferably but not limited to the Nd-YAG pulse laser is used thereto.

In one embodiment of the present invention it shows that the ultrasound generated in the built-in sensor 2 for measuring (31). Built-in sensor 31 as having a very high sensitivity and a high SNR (signal-to-noise ratio) to measure the ultrasonic wave, preferably a piezoelectric sensor or a high sensitivity but is not limited to this. The use of this sensor will be used and power / Because cable installation is not required for data transmission has a cable twist rotation conductive liquid (mercury), the slip ring so that the electrical signal is transmitted through the (slip ring). Other embodiments of the piezoelectric sensor as a wireless sensor node that does not use the cable.

Using non-contact meter (32, a laser interferometer) in place of the built-in sensor, as shown in another embodiment of the sensor 3 is measured as the ultrasound. Examples of non-contact laser vibrometer is the instrument by, Lee, Kwang-wave mixing interferometer (TWM-PI, Two Wave Mixing Photorefractive Interferometer), Fabry-Perot interferometer, etc. (CFPI, Confocal Fabry-Perot Interferometer).

In general, a laser interferometer system for measuring ultrasonic surface displacements to proceed from the structure and the structure surface by measuring the phase shift of light by the surface of the displacement structure. Laser vibrometer 32 is a modified example of the laser interferometer, as is the structure and the surface irradiated with the laser beam by measuring a wavelength variation of the laser beam reflected by the Doppler effect to measure the surface velocity of the ultrasonic measurement is also possible structures. Lee, Kwang-wave mixing interferometer is possible to remove the low-frequency signal by using the photorefractive medium in the process and the structure surface displacement measurement device only measures the high-frequency signal. Fabry-Perot interferometer is a device that compares the wavelength change of the laser beam reflected by the Doppler effect and the specific resonant wavelengths of the interferometer to measure the surface velocity of the structure. The non-contact laser interferometer 32 can be measured efficiently without affecting the target structure, such as it is not necessary to provide the built-in sensor unlike the measurement points, but also free to be able to determine a non-contact structure to the sensor / cable.

4 is a structural diagram of a single, integrated system integrity assessment, ultrasound generating system 10, the laser pulse control system 20 is configured by an ultrasonic data acquisition system 30 and the damage detection system 40.

Ultrasound generating system 10 includes a galvanometer for the reorientation of the irradiating laser beam to the desired position on one side of a system for generating an ultrasonic wave to irradiate the laser beam at the rotating structure of the blade (1), the blade (1) Use meter 14.

Pulsed laser control system 20 is composed of a positioning portion 21 and the angle sensor 22 on the main computer. Angle sensor 22 is installed on the rotation shaft of pulse laser structure 11 and to synchronize the blade (1) by irradiating a laser beam only enter in the target range to prevent damage to the person or the animal with a laser beam. To use optical encoders using a sensor (angle sensor) and the blade (1) to generate an electrical pulse signal each time one revolution is used to detect the initial position.

The ultrasonic measurement system 30 includes a digitizer for collecting and storing signals from the ultrasonic sensor and the sensor for detecting the ultrasonic wave generated by the installation to the other side of the laser irradiation point. The ultrasonic sensor is used in a non-contact instrument such as a built-in sensor and a laser oscillating system mentioned above.

The damage detection system 40 is composed of an ultrasonic image processing unit 41 and the automatic corruption detection section 42. Damage detection system 40 is detected by imaging the collected data, the damage is that when a process change aspects of ultrasound to visualize and extract only the corrupted information via the image processing, a maintenance manager or the like through the initial and warning alarm It informs to the.

In a preferred embodiment of the present invention, the damage detection system 40 from the generated ultrasound ultrasonic image-signal to extract the damaged interaction information / image processing techniques, the extraction ultrasonic visualization of the damage to the damaged interaction information position and damage visualization technique that enables to determine the severity of the intuitively, the extracted ultrasound-automatically the presence of the damage and its position, and in that it comprises an automatic damage diagnostic techniques that provide information about the severity of liver injury from interaction information It characterized.

Generating position of the ultrasound to the shape of the blade, influenced of vibration and wind station and confirms the position of the ultrasonic waves generated by the impact to control localization techniques.

Figure 5 is illustrated as a step-by-step technique to determine the impact of the imaging position by using a correlation between the ultrasonic signals and the generated data is a technique to identify and study the generation position. The first step is to investigate the method comprising: collecting training data from a stationary laser pulse on one single point of the blade and collects the ultrasound generated using the built-in sensor and a laser oscillating system. Wherein repeatedly over an area around the point to create a training data set of learning the laser is irradiated to create a training data set and then scanning collects ultrasound generated for all study points. The second step is to collect, store, and ultrasonic irradiation of

a laser beam generated in a rotating state by the sensor. Again, by irradiating a laser beam repeatedly around the zone with respect to the irradiation zone to create the image area. Using techniques to evaluate the exact impact position occurred in the measured signal. The third step estimates the position by analyzing the correlation between the measured data of the learning data and the image rotation and imaging.

Impact location techniques as to correlate with the training data with the measured signals, the generation and detection of ultrasonic apparatus (mechanism) is a correlation index of the learning data and the measured signal is the same is the maximum at the same position. Therefore, the learning point with the maximum correlation index and the measured signal can be identified as the most likely point of an ultrasonic generator.

Is called the measured signal $f(t)$ and correlation index of the two signals to said learning signal $g(t)$ is expressed as equation (1).

Equation 1

$$(f * g)(\tau) = \int_{-\infty}^{+\infty} f(t)g(t+\tau)dt$$

Wherein "*" represents a correlation operator. Equation (1) it is catch much calculation time due to the integration with respect to time domain. In order to reduce the computation time on the basis of the convolution theorem (convolution theorem) and the Fourier transformation is calculated by a Fourier transform and the inverse Fourier function, as shown in equation (2). Sign in the second equation is a convolution operator.

Equation 2

$$f * g = f(-t) \otimes g = F^{-1} \{ Ff(-t) \cdot F(g) \}$$

Image processing technique is used in the opposite principle (reciprocal theorem). That is, the measured wave using the principle is identical to that measured at the generation point to generate an ultrasonic wave in the measuring point, and examining the various points in the laser and measuring the ultrasonic wave from one fixed point to the sensor, which is fixed the ultrasonic source (ultrasonic wave source), and the same as measured for the desired wave generated in the spatial domain. By imaging the spatial information may indicates a time for the ultrasonic imaging process sheet.

Damage visualization techniques may calculate the wave energy at each point from the disciplined ultrasound information, this information can be represented for the entire time domain through a single image. Ultrasonic who are going to form a standing wave is trapped within the damage zone, have a higher pulse energies than the other point is due. Therefore, the wave energy can be found by finding the location of the damage to a particularly high region. In the present invention, using the standing wave filter (standing wave filter) to highlight the damaged area. Standing wave filters, only the image crystallized the standing wave technique to extract information from an ultrasonic measuring chapter in view of the fact that the standing wave is formed in the damaged area.

Automatic damage diagnosis technique to inform the manager in a display or alarm, or communication means such as SMS information about whether and to automatically position the presence and severity of the damage as compared to the reference value by calculating the energy value of the trapped wave standing wave to the damage area a technique.

6 has been made to show the experimental example according to the present invention, (a) an average of 10 laser irradiated to the conference, and generate ultrasound signals for each image generated at the stationary point is blue for a red signal of a positive color refers to the level of the signal is negative. Ultrasonic wave goes round because there is no damage to reputation. (B) is as a ultrasound imaging using conventional methods inde ultrasound image of a blade rotating at 20 rpm, the image is damaged by vibration and the rotation variation of the time delay between the laser pulse and the encoder can not know the progression of the ultrasonic . But (c) is correct, the analysis of the correlation between the training data and the measured signal image technique in accordance with the present invention, and shows a clear ultrasonic process.

Figure 7 is an ultrasound image processing result according to the present invention, showing the process aspect of the ultrasound is changed when there is damage to the blade.

Figure 8 shows that show that the visualization by extracting only damage the imaging information as the damage visualization technique in accordance with the present invention can monitor the site of injury that is intuitive and effective.

The present invention is to provide visualization of the location of the damaged interaction, size, and information about the severity of the damage and automatically between the ultrasonic process, the turbine blades of a wind turbine, helicopters, aircraft and marine engine rotating a propeller, such as a variety of structures It can be monitored for quality.

PATENT CITATIONS

Cited Patent	Filing date	Publication date	Applicant	Title
JPH1172591A *				<i>Title not available</i>
KR20020072314A *				<i>Title not available</i>
KR20100079238A *				<i>Title not available</i>
KR20110098301A *				<i>Title not available</i>

* Cited by examiner

NON-PATENT CITATIONS

Reference

1	*	LEE, J. R. ET AL.: 'Long distance laser ultrasonic propagation imaging system for damage visualization.' OPTICS AND LASERS IN ENGINEERING. vol. 49, August 2011, pages 1361 - 1371
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* Cited by examiner

CLASSIFICATIONS

International Classification	G01B17/04 , G01N29/04
Cooperative Classification	G01N2291/0289 , G01N2291/023 , G01N29/04 , G01N29/043 , G01M7/00 , G01N29/2418 , G01N2291/2634 , G01N29/069 , G01N2291/2693

LEGAL EVENTS

Date	Code	Event	Description
Mar 26, 2014	121	Ep: the epo has been informed by wipo that ep was designated in this application	Ref document number: 13824795 Country of ref document: EP Kind code of ref document: A1
Jan 29, 2015	WWE	Wipo information: entry into national phase	Ref document number: 14418427 Country of ref document: US
Feb 2, 2015	NENP	Non-entry into the national phase in:	Ref country code: DE

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